Definitions

Johan Voskamp

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1

Definition 1.1 (No-lonely-color graphs)**.** A graph G is no-lonely-colored if there exists an edge-coloring such that

- Every vertex's edges can only have at most two of the same color.
- Every cycle contains no color exactly once.

Definition 1.2 (Cayley Graphs)**.** Let G be a group, and let $S \subseteq G$, construct the graph $C(G, S)$ as follows:

- every $g \in G$ is a vertex,
- for every ordered pair (g, s), where $s \cdot g_1 = g_2$ with $s \in S$ and $g_1, g_2 \in G$, set an edge between g_1 and g_2, disallowing identical edges.

The resulting object, $C(G, S)$ is then called a Cayley Graph.

Definition 1.3 (Minimal Cayley Graphs)**.** A Cayley Graph $C(G, S)$ is called minimal when S is a smallest possible subset still generating the group G.

Definition 1.4 (Girth)**.** The Girth of a graph is the length of its shortest cycle, if no cycles exist the girth is set to be ∞.

Definition 1.5 (Multigraphs)**.** A Multigraph is a graph where it is allowed for two vertices to have multiple edges connecting each other.

Definition 1.6 (Bridges)**.** Let M be a Multigraph and e be an edge of that graph. If removing e from M results in a graph that has more connected components, we call e a bridge.

Definition 1.7 (Hypergraphs)**.** Let V be a set of elements and let $E \subset P(X)$. We call the collection (V, E), an (undirected) Hypergraph. The subset E is the collection of hyperedges of the Hypergraph.

Definition 1.8 (Chromatic Number)**.** The chromatic number of a (hyper)graph G is the smallest possible number of colors necessary to color the vertices G if adjacent vertices can't have the same color.

Definition 1.9 (r-uniform Hypergraphs)**.** Let (V, E) be a Hypergraph. If each Hyperedge of the Hypergraph contains exactly k elements, we call the Hypergraph k-uniform.

Definition 1.10 (Tranquil Hypergraphs)**.** An r-uniform hypergraph is tranquil if there exists a set of the same labels that is applied on every hyperedge, it holds that for every possible closed walk in the hypergraph, when projected as a multigraph from label to label, there are no bridges.

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